

Lumber, Masonry and What They Are Sold By

The naming questions. Every one of these is a plain-term answer, which the signal words tell you in advance.

S4S	Lumber that has been planed on all sides — surfaced four sides.
Board foot	The unit dimensional framing lumber is sold by. One board foot is 12 in by 12 in by 1 in: one square foot, one inch thick. It measures volume, which is why lumber pricing does not follow the nominal size on the label.
Deformed bars	The other name for rebar. The deformations are the ridges rolled into the surface, and they are what lets concrete grip the steel.
Bonding strength	The most important property for mortar. Mortar is not there to carry load — the masonry units do that. It is there to stick them together and seal between them.

YOU HAVE MISSED THIS 3 TIMES

Lumber that has been planed on all sides is called:

S4S.

YOU HAVE MISSED THIS 2 TIMES

Another name for rebar is:

Deformed bars.

YOU HAVE MISSED THIS

Dimensional framing lumber is sold by the:

Board foot.

YOU HAVE MISSED THIS

What is the most important property for mortar?

Bonding strength.

Nominal size against actual size

Both lumber and masonry are named by a size they are not. The name is the size before finishing; the actual piece is smaller. For masonry the shortfall is three-eighths of an inch in each direction, which is the mortar joint — the nominal size is the block plus its joint, so a wall built of them comes out at a round number.

The numbers here

Masonry block — actual against nominal

minus 3/8 in each way

An 8 x 8 x 16 block, actually

7.625 x 7.625 x 15.625 in

YOU HAVE MISSED THIS 2 TIMES

The actual size of an 8 x 8 x 16 masonry block is:

7.625 x 7.625 x 15.625 inches.

Keeping water out of a wall below grade

In walls above grade, where there is no hydrostatic pressure, the protection used is damp-proofing. Damp-proofing resists moisture moving through by capillary action. Waterproofing is a membrane built to hold back liquid water under pressure, and you only need it where a high water table creates that pressure.

Say it back

Cover the page. Answer each one out loud.

1. What does S4S stand for, and what is a board foot?
2. Give the other name for rebar, and say what the deformations do.
3. What is the most important property of mortar, and why is it not strength?
4. Actual dimensions of an 8 x 8 x 16 block, and where the missing 3/8 goes.
5. Damp-proofing or waterproofing above grade, and what decides it?